

Impact Statement

Yasra Chandio
 [yasrachandio.github.io](https://github.com/yasrachandio)

My approach to my academic career is fundamentally shaped by my personal and professional experiences, and reinforced through my academic and outreach activities. I believe in creating an environment where people from all backgrounds feel valued, supported, and empowered to succeed. Through my teaching, research, and service, I strive to reduce barriers, build community, and promote equitable opportunities in academia and beyond.

Personal Background and Academic Service. As a female researcher from a developing country and a first-generation woman pursuing engineering and earning a graduate degree in the USA, I have navigated significant academic and cultural challenges. These experiences give me a unique understanding of the barriers that hinder student success and fuel my commitment to building supportive communities in which all students can thrive. This experience has given me a firsthand, unique understanding of the academic and cultural barriers that can hinder student success, and it has motivated me to build a supportive environment that enables all students to thrive.

At the college level, I chair the College of Engineering (CoE) Dean's Graduate Advisory Group, where I translate student feedback into high-impact programs and timely solutions. A key achievement was organizing the inaugural CoE Graduate Research Summit, which featured 50 posters and 10 lightning talks, creating a critical platform for graduate research visibility. I recruited and coordinated 25 faculty reviewers to ensure a rigorous selection process. Building on this success, I launched the CoE Grad Life Series, offering workshops and panels on research ethics, job market preparation, and work-life balance. I also led the efforts to allot dedicated TA rooms. Through these efforts, I both advocate for student interests and actively build resources and community for my peers.

In my department, I co-organize the ECE Women Dinner Series, supported by a competitive campus climate grant, and was the first student inductee into the ECE DEI Committee. I helped launch a Tea Time series for informal peer discussions, co-developed a peer advising program pairing freshmen with upper-level mentors, and founded the ECE graduate gender diversity group. Growing from three to over thirty members, the group now holds bi-weekly meetings and retreats, funded through grants I secured, to provide mentorship, community, and professional development for female and non-binary graduate students. I have also advocated for students in crisis, including working with the Dean's office to resolve a conflict over shared lab spaces. My efforts resulted in a new college-wide policy that guarantees equitable access to shared labs. In recognition of these contributions, I received the 2024 UMass CoE DEI Award.

Beyond my institution, I am committed to outreach and mentorship that expands access to STEM. I designed and led a three-week embedded systems course for Springfield High School, introducing seniors from many underrepresented communities to programming and hardware integration. I have mentored students in the UMass Turing Summer Program through lectures and research demos, and I run sessions with elementary school and high school robotics clubs to spark curiosity about emerging technologies. These initiatives aim to make computing accessible to students who might not otherwise have the opportunity to encounter it. I am dedicated to outreach and mentorship that inspires the next generation of leaders.

Research and Broader Impact. *In my research,* I focus on designing safer, more adaptive Mixed Reality (MR) systems that provide secure, fair, and personalized experiences. My work addresses pressing issues such as bias, safety risks, and equitable system design. *In the broader research community,* I have advanced DEI through national roles. As a CRA-E Grad Fellow and editor of the CRA Undergraduate Research Highlights, I spotlighted the work of students from diverse backgrounds, sharing their unique journeys and contributions. I have mentored underrepresented undergraduates through the NSF REU Computing for an Equitable Energy Transition (CEET) program, guiding them to tackle technical problems with real-world societal impact.

Future Plans

I am committed to advancing inclusive institutional change through mentoring, academia-industry partnerships, inclusive networking, and sustained K-12 outreach. In parallel, my research prioritizes equitable, sustainable, and open human-centered technologies that address bias and accessibility. By extending these efforts beyond my lab, I aim to support long-term professional growth and broaden participation across academia and the wider STEM ecosystem.